

HIV & Child Mortality in Africa

A data-driven analysis of HIV prevalence and child mortality trends across East African Community nations, revealing critical insights for public health interventions.



Global HIV Burden: The 75% Reality

75%

Global Concentration

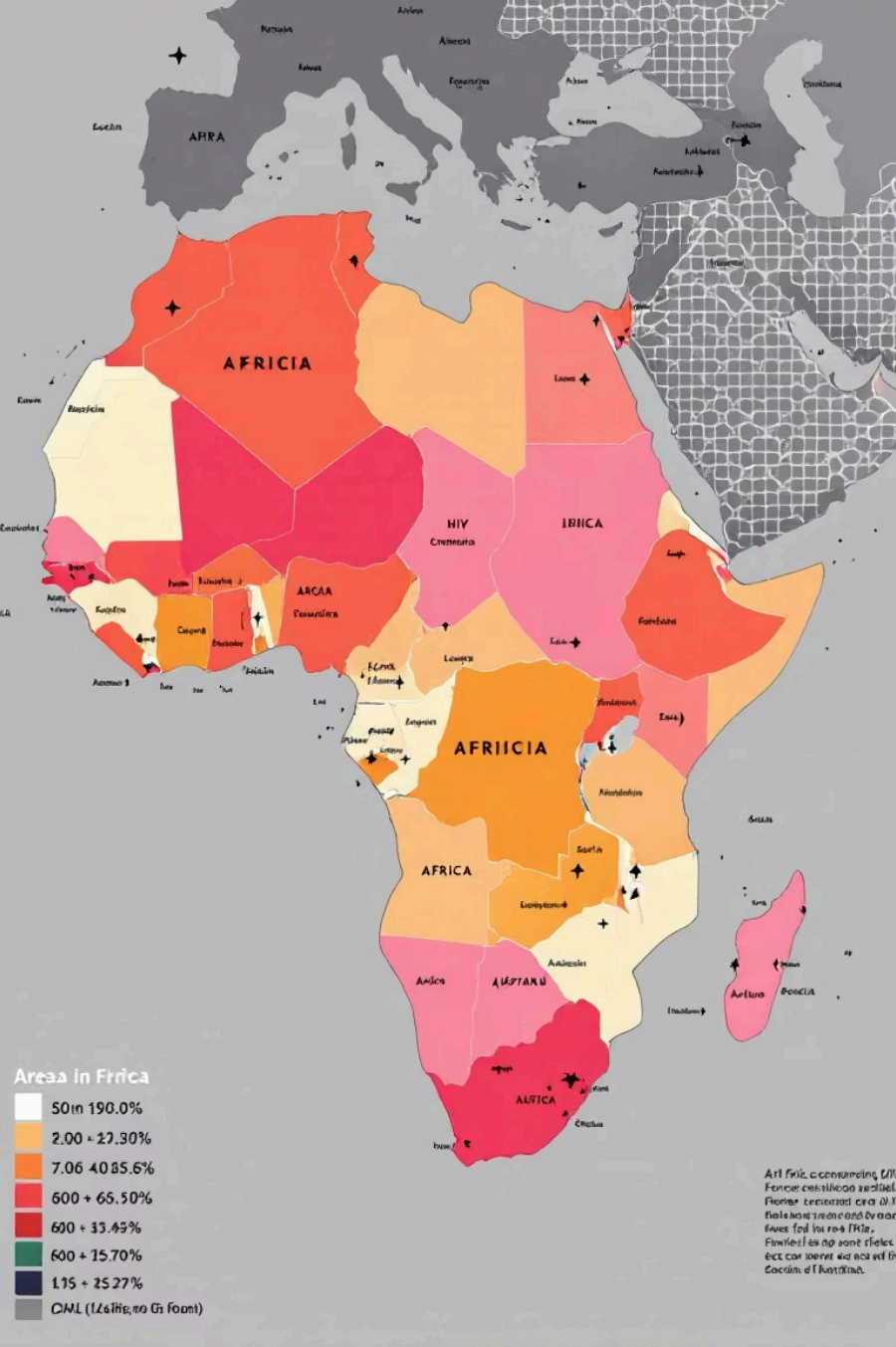
Just 15 countries account for three-quarters of all people living with HIV worldwide.

85%

Africa's Share
Sub-Saharan Africa bears the overwhelming majority of this concentrated burden.

21%

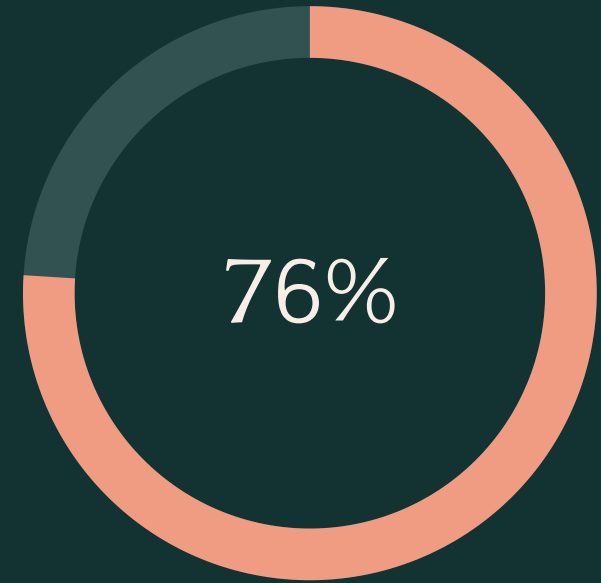
South Africa Leads
South Africa alone contributes over one-fifth of the global HIV burden with 6.4M cases.



Top 15 Countries: A Critical Focus

These nations represent the epicenter of the global HIV epidemic. South Africa, India, Mozambique, Nigeria, and Kenya lead the list, each facing unique challenges in prevention and treatment.

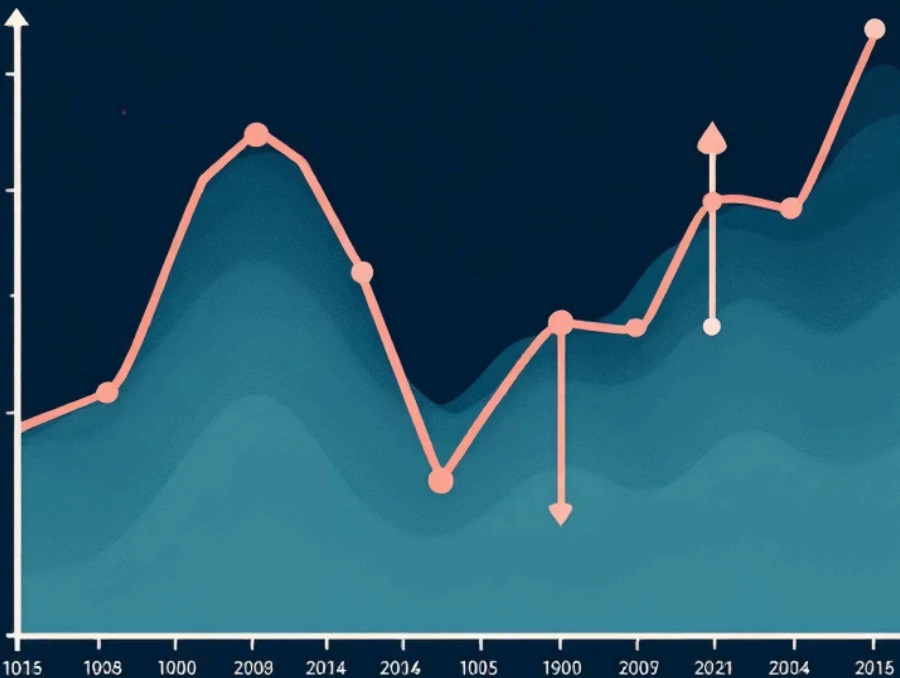
- South Africa: 6.4M cases (20.9% of global burden)
- India: 2.5M cases (8.1%)
- Mozambique: 1.9M cases (6.2%)
- Nigeria: 1.9M cases (6.1%)
- Kenya: 1.5M cases (4.9%)



Combined Impact

These 15 countries together account for over three-quarters of all HIV cases globally.

Diverging Trajectories: Success & Struggle



Declining Cases



Kenya, Ethiopia, Zimbabwe, and DRC saw steady declines through aggressive ART rollout and prevention campaigns.

Rising Cases



South Africa, Mozambique, and Nigeria continue reporting increases due to improved survival and population growth.

Mixed Patterns



Malawi and Zimbabwe show volatile trends reflecting economic disruptions and resource constraints.

Regional Patterns Across WHO Zones

Africa: 85% of Burden

Twelve African nations dominate the list, with South Africa's epidemic continuing to grow despite treatment advances.

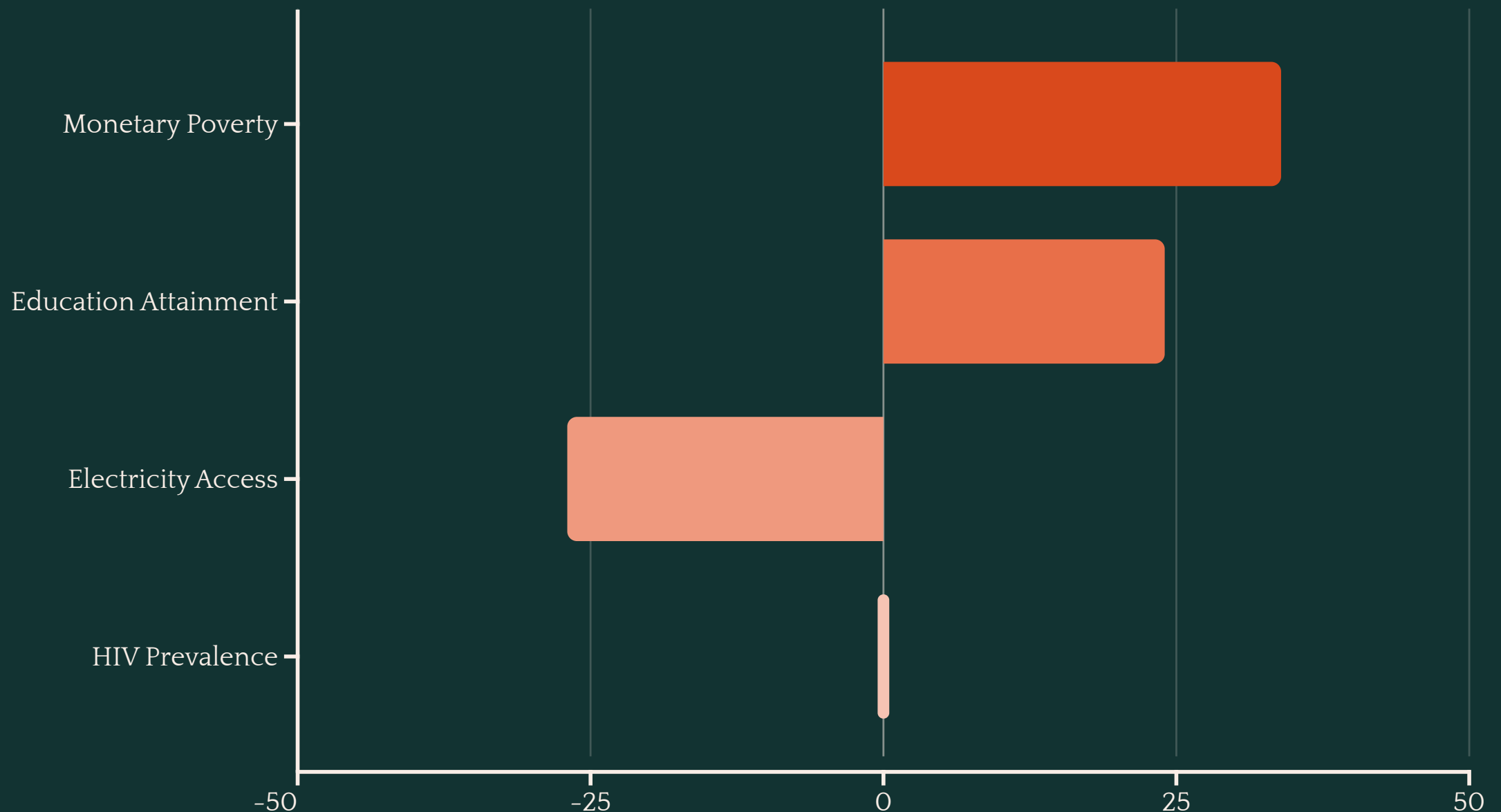
South-East Asia: 8%

India leads with 2.5M cases; Thailand shows remarkable decline from 920K to 580K through early intervention.

Americas: 7%

Brazil's epidemic grows steadily, reaching 1M cases by 2023, driven by urban concentration and demographic shifts.

The Poverty Connection



Statistical modeling reveals monetary poverty and education drive multidimensional poverty, while electricity access reduces it significantly. Surprisingly, HIV prevalence shows no direct statistical relationship—suggesting poverty's roots lie in broader structural inequalities.

Correlation coefficient: 0.73 between poverty headcount ratio and HIV infections indicates strong association, but causality flows through economic pathways.



Child Mortality in East Africa

104

Somalia: Highest
U5MR

Per 1,000 live births

40

South Sudan:
Highest NMR

Per 1,000 live births

18

Rwanda: Lowest
NMR

Remarkable progress
since 2010

Neonatal and under-five mortality rates vary dramatically across the East African Community, reflecting differences in healthcare investment, political stability, and public health priorities.

Rwanda's Remarkable Transformation



From Crisis to Leadership

Rwanda now boasts the lowest child mortality rates in East Africa, overtaking Kenya's long-held position. This transformation stems from:

- Community-based health insurance (Mutuelle de Santé)
- Massive investment in rural health infrastructure
- Strong political commitment to health equity
- Integration of maternal-child health services

Key Findings & Implications

01

Geographic Concentration

HIV burden concentrates in 15 countries; targeted interventions in these nations could dramatically impact global prevalence.

02

Electricity as Catalyst

Access to electricity shows strongest negative correlation with poverty –infrastructure investment yields health dividends.

03

Conflict's Toll

Somalia and South Sudan's high mortality rates underscore how political instability undermines health systems.

04

Success is Possible

Rwanda and Thailand demonstrate that sustained commitment and community-based approaches can transform outcomes.





Recommendations for Action



Expand Infrastructure

Prioritize electricity access in high-burden regions as a poverty-reduction strategy with health co-benefits.



Strengthen Health Systems

Invest in community-based care models proven effective in Rwanda and Thailand.



Focus Resources

Concentrate prevention and treatment efforts in the 15 countries bearing 75% of global HIV burden.



Cross-Border Collaboration

Enhance regional cooperation in East Africa to share best practices and coordinate responses.